UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

ALGEBRAIC STRUCTURES

LECTURE NO: 34

Algebraic Structure

The algebraic structure is a type of non-empty set G which is equipped with one or more than one binary operation. Let us assume that * describes the binary operation on non-empty set G. In this case, (G, *) will be known as the algebraic structure. (1, -), (1, +), (N, *) all are algebraic structures.

(R, +, .) is a type of algebraic structure, which is equipped with two operations (+ and .)

Binary Operation of Set

In the binary operation, binary stands for two. A binary operation is a type of operation that needs two inputs, which are known as the operands. When we perform multiplication, division, addition, or subtraction operations on two numbers, then we will get a number. The two elements of a set are associated with binary operations. The result of these two elements will also be in the same set. So we can say that if we perform a binary operation on a set, then it will perform calculations that combine two elements of the set and generate another element that belongs to the same set.

Let us assume that there is a non-empty set called G. A function f from G × G to G is known as the binary operation on G. So f: G × G → G defines a binary operation on G.

Examples of Binary operation

In this example, we will take the two natural numbers or two real numbers and perform binary operations such as addition, multiplication, subtraction, and division on these numbers. The algebraic operation on two natural numbers or real numbers will generate a result. If we get a natural number or real number as a result, then we will consider that binary operation in our set.

Addition:

We will learn about addition, which is a binary operation. Suppose we have two natural numbers(a, b). Now if we add these numbers, then it will generate a natural number as a result. **For example:** Suppose there are 6 and 8 two natural numbers and the addition of these numbers are

6 + 8 = 14

Hence, the result 14 is also a natural number. So, we will consider an addition in our set. The same process will be followed for real numbers as well.

+: N + N → N is derived by (a, b) → a + b
+: R + R → R is derived by (a, b) → a + b

Multiplication:

Now we will learn multiplication, which is a binary operation. If we multiply two natural numbers (a, b), then it will generate a natural number as a result. **For example:** Suppose there are 10 and 5 two natural numbers and the multiplication of these numbers are:

10 * 5 = 50

Hence, the result 50 is also a natural number. So we will consider multiplication in our set. The same process will be followed for real numbers as well.

+: N × N → N is derived by (a, b) → a × b
+: R × R → R is derived by (a, b) → a × b

Subtraction:

Now we will learn subtraction, which is a binary operation. If we subtract two real numbers (a, b), then it will also generate a real number as a result. The same process will not be followed for natural numbers, because if we take two natural numbers to perform binary subtraction, then it is not compulsory that it will generate a natural number. **For example:** Suppose we take two natural numbers 5 and 7 and the subtraction of these numbers are

5 - 7 = -2

Hence, the result is not a natural number. So we will not consider subtraction in our set.

- : R x R → R is derived by (a, b)→ a - b

Division

Now we will learn division, which is a binary operation. If we divide two real numbers (a, b), then it will also generate a real number as a result. The same process will not be followed for natural numbers, because if we take two natural numbers to perform binary division, then it is not compulsory that it will generate a natural number. **For example:** Suppose we take two natural numbers 10 and 6 and the division of these numbers is

$$10/6 = 5/3$$

Hence, the result 5/3 is not a natural number. So we will not consider division in our set.

- : R - R → R is derived by (x, y) → x - y

Properties of Algebraic structure

Commutative: Suppose set G contains a binary operation *. The operation * is called to be commutative in G if it holds the following relation:

$$x * y = y * x \text{ for all } x, y \text{ in } G$$

Associative: Suppose set G contains a binary operation *. The operation * is called to be associative in G if it holds the following relation:

$$(x*y)*z = x *(y*z) \text{ for all } x, y, z \text{ in } G$$

Identity: Suppose we have an algebraic system (G, *) and set G contains an element e. That element will be called an identifying element of the set if it contains the following relation:

$$x * e = e * x = x \text{ for all } x$$

Here, **element e** can be referred to as an identity element of G, and we can also see that it is necessarily unique.

Inverse: Suppose there is an algebraic system (G, *), and it contains an identity e. We will also assume that the set G contains the elements x and y. The element y will be called an inverse of x if it satisfies the following relation:

$$x * y = y * x = e$$

Here, **element x** can also be referred to as inverse of y, and we can also see that it is necessarily unique. The inverse of x can also be referred to as x^{-1} like this:

$$x * x^{-1} = x^{-1} * x = e$$

Cancellation Law: Suppose set G contains a binary operation *. The operation * is called to be left cancellation law in G if it holds the following relation:

$$x * y = x * z \text{ implies } y = z$$

It will be called the right cancellation law if it holds the following relation:

$$y * x = z * x \text{ implies } y = z$$

Types of Algebraic structure

There are various types of algebraic structure, which is described as follows:

- Semigroup
- Monoid
- Group
- Abelian Group

All these algebraic structures have wide application in particular to binary coding and in many other disciplines.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

TYPES OF ALGEBRAIC STRUCTURES

LECTURE NO: 35

Types of Algebraic Structure

There are various types of algebraic structure, which is described as follows:

- Semigroup
- Monoid
- Group
- Abelian Group

All these algebraic structures have wide application in particular to binary coding and in many other disciplines.

Semi Group

Suppose there is an algebraic structure (G, *), which will be known as semigroup if it satisfies the following condition:

- **Closure:** The operation * is a closed operation on G that means (a*b) belongs to set G for all a, b ∈
- **Associative:** The operation * shows an association operation between a, b, and c that means a*(b*c) = (a*b)*c for all a, b, c in G.

Note: An algebraic structure is always shown by semigroup.

Example 1:

The examples of semigroup are (Matrix, *) and (Set of integer, +).

Example 2:

The semigroup contains a set of positive integers with an additional or multiplication operation. The positive integers will not contain zero. **For example:** Suppose we have a set G, which contains some positive integers except zero such as 1, 2, 3, and so on like this:

G = {1, 2, 3, 4, 5, ...}

- This set contains the closure property because according to closure property (a * b) belongs to G for every element a, b. So in this set, (1*2) = 2 ∈

- This set also contains the associative property because according to associative property (a + b) + c = a + (b + c) belongs to G for every element a, b, c. So in this set, (1 + 2) + 3 = 1 + (2 + 3) = 6 ∈

Monoid:

A monoid is a semigroup, but it contains an extra **identity element** (E or e). An algebraic structure (G, *) will be known as a monoid if it satisfies the following condition:

- **Closure:** G is closed under operation * that means (a*b) belongs to set G for all a, b ∈
- **Associative:** Operation * shows an association operation between a, b, and c that means a*(b*c) = (a*b)*c for all a, b, c in G.
- **Identity Element:** There must be an identity in set G that means a * e = e * a = a for all x.

Note: An algebraic structure and a semigroup are always shown by a monoid.

Example 1:

In this example, we will take (Set of integers, *), (Set of natural numbers, +), and (Set of whole numbers, +). Where

- Monoid is shown by (Set of Integers, *) because 1 is an integer and it is also an identity element.
- Monoid is not shown by (Set of natural numbers, +) because there is not an identity element, but it is a semigroup.
- Monoid is shown by (Set of whole numbers, +) because it contains 0 as the identity element.

Example 2:

The monoid contains a set of positive integers with additional or multiplication operations except zero. **For example:** Suppose we have a set G, which contains some positive integers like 1, 2, 3, and so on like this:

G = {1, 2, 3, 4, 5, …..}

- This set contains the **closure property** because according to closure property (a * b) belongs to G for every element a, b. So in this set, (1*2) = 2 and so on.
- This set contains the **associative property** because according to associative property (a + b) + c = a + (b + c) belongs to G for every element a, b, c. So in this set, (1 + 2) + 3 = 1 + (2 + 3) = 5, and so on.
- This set also contains the **identity property** because according to this property a * e = e * a = a, where a ∈ So in this set, (2 × 1) = 2, (3 × 1) = 3, and so on. In our case, 1 is the identity element.

Group:

A Group is a monoid, but it contains an extra **inverse element**, which is denoted by 1. An algebraic structure (G, *) will be known as a group if it satisfies the following condition:

- **Closure:** G is closed under operation * that means (a*b) belongs to set G for all a, b ∈
- **Associative:** * shows an association operation between a, b, and c that means a*(b*c) = (a*b)*c for all a, b, c in G.
- **Identity Element:** There must be an identity in set G that means a * e = e * a = a for all a.
- **Inverse Element:** It contains an inverse element that means a * a^{-1}= a^{-1} * a = e for a ∈

Note: An algebraic structure, semigroup, and monoid are always shown by a Group.

Example 1:

The examples of group are Matrix multiplication and (Z, +).

Example 2:

In this example, we will use the matrix multiplication operation on the set of non-singular matrices N × N from a group.

- If we perform multiplication of non-singular matrices N × N, then it will also be a non-singular matrix N × N, which holds the property of **closure**.
- Matrix multiplication itself holds the property of **association**. So it is also associative.
- The identity matrix is contained in the set of non-singular matrices N × N, which holds the property of **identity element**.
- As we have seen that all the matrices are non-singular. So they will contain the inverse elements, which will be also non-singular matrices. Hence, it also holds the property of **inverse**.

Abelian Group

An abelian group is a group, but it contains **commutative law**. An algebraic structure (G, *) will be known as an abelian group if it satisfies the following condition:

- **Closure:** G is closed under operation * that means (a*b) belongs to set G for all a, b ∈
- **Associative:** * shows an association operation between a, b, and c that means a*(b*c) = (a*b)*c for all a, b, c in G.
- **Identity Element:** There must be an identity in set G that means a * e = e * a = a for all a.
- **Inverse Element:** It contains an inverse element that means a * a^{-1}= a^{-1} * a = e for a ∈
- **Commutative Law:** There will be a commutative law such that a * b = b * a such that a, b belongs to G.

Note: (Z, +) is an Abelian group because it is commutative, but matrix multiplication is not commutative that's why it is not an abelian group.

Example: Suppose we have a set G, which contains some positive integers except zero such as 1, 2, 3, and so on with additional operations like this:

G = {1, 2, 3, 4, 5, ...}

- This set contains the **closure property** because according to closure property (a + b) belongs to G for every element a, b. So in this set, (1 + 2) = 2 ∈ G and so on.
- This set also contains the **associative property** because according to associative property (a + b) + c = a + (b + c) belongs to G for every element a, b, c. So in this set, (1 + 2) + 3 = 1 + (2 + 3) = 6 ∈ G and so on.
- This set also contains the **identity property** because according to this property (a * e) = a, where a ∈ So in this set, (2 × 1) = 2, (3 × 1) = 3, and so on. In our case, 1 is the identity element.
- This set also contains the **commutative property** because according to this property (a * b) = (b * a), where a, b ∈ So in this set, (2 × 3) = (3 × 2) = 6 and so on

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

MITRC

UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

SUBGROUPS

LECTURE NO: 36

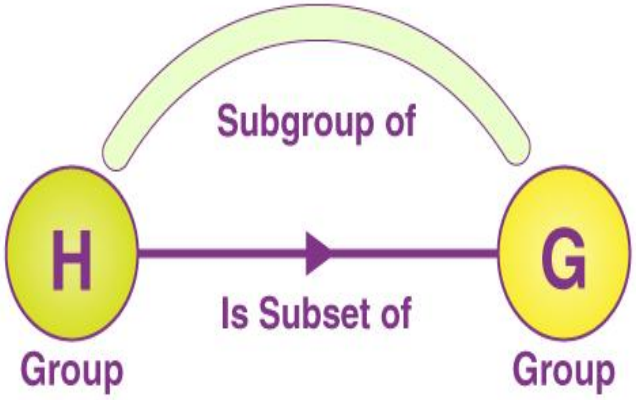
Subgroups

Definition

Let (G, ⋆) be a group and H be a non-empty subset of G, such that (H, ⋆) is a group then, "H" is called a subgroup of G.

That means H also forms a group under a binary operation, i.e., (H, ⋆) is a group.

Also, any subset of a group G is called a complex of G.



Below are some important points about subgroups.

- A subset H of a group G is a subgroup of G, if H itself is a group under the operation in G.
- A subgroup of a group consisting of only the identity element, i.e., {e} is called the trivial subgroup.
- A subgroup H of a group G, a proper subset of G, i.e., H ≠ G is called the proper subgroup and is represented by H < G. This can be read as "H is a proper subgroup of G".
- If H is a subgroup of G, then G may be called an over group of H in some cases.

Theorems on Subgroups

Theorem 1:

H is a subgroup of G. Prove that the identity element of H is equal to the identity element in G.

Proof:

Given that H is a subgroup of G.

Let us assume that e and e' be the two identity elements in H and G, respectively.

Let $a \in H \Rightarrow a \in G$ [since H is a subset of G]

Identity element in group H = e

Thus, $a \star e = e \star a = a$.....(1)

Identity element in group G = e

Therefore, $a \star e' = e' \star a = a$.....(2)

From (1) and (2),

$a \star e = a \star e'$

$\Rightarrow e = e'$

That means, the identity element in H is equal to the identity element in G.

Hence proved.

Theorem 2:

H is a subgroup of G. The inverse of any element in H is equal to the inverse of the same element in G.

Proof:

Given that H is a subgroup of G.

Consider $a \in H \Rightarrow a \in G$

Let us assume that b and c are two inverse elements of a in H and G respectively.

Let b be the inverse element of a in H.

Then, $a \star b = b \star a = e$....(1)

Let c be the inverse element of a in G.

Then, $a \star c = c \star a = e$....(2)

From (1) and (2),

$a \star b = a \star c$

$\Rightarrow b = c$

That means the inverse element of a in H is equal to the inverse element of a in G.

Hence proved.

Difference between Groups and Subgroups

The below table illustrates a few differences between groups and subgroups.

Group	Subgroup

A group is a set combined with a binary operation, such that it connects any two elements of a set to produce a third element, provided certain axioms are followed.	A subgroup is a subset of a group. H is a subgroup of a group G if it is a subset of G, and follows all axioms that are required to form a group.
Groups satisfy the following laws: • Closure • Associative • Identity element • Inverse law	Subgroups also satisfy the following laws: • Closure • Associative • Identity element • Inverse law
The number of elements of a finite group is called the order of a group.	A subgroup is also a group, and the order of a subgroup is less than the order of a group.

Properties of Subgroups

We can also prove the following statements using the properties of groups and subgroups.

1. Let H be any subgroup of G, such that H-1 = H and HH = H.
2. H is a non-empty complex of a group G. The necessary and sufficient condition for H to be a subgroup of G is: a, b $\in$ H $\Rightarrow$ ab-1 $\in$ H, where b-1 is the inverse of b in G.
3. H is a subgroup of G if and only if HH-1 = H.
4. If H and K are two subgroups of a group G, then HK is a subgroup of G if and only if HK = KH.
5. If H and K are two subgroups of a group G, then H $\cap$ K is a subgroup of G.
6. The union of two subgroups of a group is a subgroup, if and only if one is contained in the other. (or) If H and G are two subgroups of G, then H U K is a subgroup of G, if and only if H $\subseteq$ K or K $\subseteq$ H.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

PROPERTIES OF GROUP

LECTURE NO: 37

Properties of Group

Order of a group

The order of a group G is the cardinality of that group. In other words, the order of a group G is the number of its elements.

Notation: The order of a group G is denoted by |G| or ◦ (G).

The following are a few examples on orders of groups.

- The group (Zn, +) is a group of order n.
- The symmetric group (S3, 0) has order 6.
- (Z, +) is a group of infinite order.

Types of Groups

Depending upon the order of groups, we can classify the groups as follows:

1. Finite Group: If a group contains a finite number of elements, then it is a finite group. For example, the symmetric group Sn is a finite group of order n!.
2. Infinite group: If a group does not have a finite number of elements, then it is an infinite group. For example, the additive group of integers is not a finite group; it is an infinite group.

Order of an element

Order of elements in a group G. The order of an element a ∈ (G, 0) is the smallest positive integer n such that an=e, where e is the identity element of G. It is denoted by |a| or ◦ (a).

If such an n exists, then we say that the element a is of finite order. Otherwise, a is said to be an element of infinite order.

Order of an Element Formula

Let a be an element of order n in a group (G, 0), that is, an=e. Then the order of ak is given by the formula:

0(ak) = ﷺgcd(ﷺ,ﷺ).

We will now understand this formula with an example.

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

Example: Let a be an element of order 20 in a group (G, 0). Find the order of a5.

Solution:

We are given 0(a)=20, that is, a20=e.

In the above formula, we have n=20 and k=5. So the order of a5 is

0(a5) = 20gcd(5,20) = 20/5 = 4.

Properties of Order of Elements in a Group

The order of an element of a group satisfies the below properties:

- The order of the identity element in a group is 1. No other element has order 1.
- Both an element and its inverse of a group have the same order. In other words, 0(a)= 0(a-1) for all elements a in G.
- Each element of a finite group has finite order and it divides the order of the group (Lagrange's Theorem). Thus no element exists in a finite group whose order exceeds the order of the group.
- If 0(a) = k and an=e, then k is a divisor of n.
- Suppose 0(a) = k. Then 0(an) = k for every integer n coprime to k.
- If 0(a) is infinite, then 0(an) is also infinite for every integer n.
- Both a and g-1ag have the same order for a, g ∈ G. These two elements are called conjugate elements of each other.
- 0(ab) = 0(ba) for all a, b ∈ G, that is, both ab and ba have the same order. Because ab and ba are conjugates of each other as ab=a(ba)a-1.

What is an Abelian Group?

A group (G, o) is called an abelian group if the group operation o is commutative. If

a o b = b o a ∀ a,b ∈ G

holds then the group (G, o) is said to be an abelian group. Abelian groups are also known as commutative groups. More specifically, if G is a non-empty set and o is a binary operation on G, then the algebraic structure (G, o) is called an abelian group if the following holds:

- The set G is closed under the operation o.
- The binary operation o is associative on G.
- G contains an identity element.
- Every element of G has an inverse in G.
- The operation o is commutative on G, that is, aob = boa ∀ a,b ∈ G.

The following are a few examples of abelian groups.

- (Z, +) is an abelian group as a+b=b+a for all a, b ∈ Z.

- The set of all rational numbers is a commutative group under the operation +.
- The group of n-th roots of unity under multiplication is an abelian (commutative) group.
- (Zn, +) is an abelian group

Non-Abelian Group

If a group G is not abelian, then G is called a non-abelian group. Non-abelian groups are also known as non-commutative groups. Examples of non-abelian groups are given below.

Non-examples of abelian groups: The symmetric group Sn; in particular S3, is not abelian. They are non-abelian or non-commutative groups.

Properties of Abelian Groups

- Every subgroup of an abelian group is abelian.
- Any cyclic group is abelian.
- Every factor (or quotient) group of a group is abelian.
- The direct product of abelian groups is also abelian.
- The center of an abelian group is abelian.
- The commutator of two elements x, y of a group G is defined by x-1y-1xy. Thus if G is an abelian group, then the commutator of any two elements of G is the identity element of G. Hence the derived subgroup of an abelian group G is trivial.
- Every group of prime order is cyclic, and hence abelian.
- If G is a group of order p2 where p is a prime number, then either $G \cong Z/2Z$ or $G \cong Z/pZ \oplus Z/pZ$. Hence a group of order p2 is abelian.
- Let p and q be two primes such that p>q and q $\nmid$ (p−1). If G is a group of order pq, then $G \cong Z/pqZ$, and hence abelian.

Examples of Abelian Groups

Question 1: Show that (Z, +) is an abelian group.

Solution:

(1) For any two integers a and b, the sum a+b is an integer. Thus Z is closed under +.

(2) We know that a+(b+c) = (a+b)+c for any a, b, c $\in$ Z. Thus the operation + is associative on Z.

(3) a+0=a for all a $\in$ Z. So 0 is an identity element in Z.

(4) As a+(-a)=0 for all a $\in$ Z, we say that -a is the inverse element of a.

(5) a+b = b+a for all a,b $\in$ Z. So + is commutative.

Thus (Z, +) is a group that is commutative. Hence (Z, +) is an abelian group.

Similarly, one can show that (Q, +), (R, +) all are abelian groups.

Definition of a Group

Let G be a set and o be a binary operation acting on it. Then the pair (G, o) is called a group if the following axioms are satisfied.

1. **[Closure]** G is closed under the composition o. That is, for every a, b ∈ G, we have that aob ∈ G.
2. **[Associativity]** o is associative. That is, for all a,b,c ∈ G, we have (aob)oc=ao(boc) ∈ G.
3. **[Identity]** There exists an element e in G such that aoe=eoa=a, for all a ∈ G.
4. **[Inverse]** For every a ∈ G, there exists a-1 ∈ G such that aoa-1 = a-1oa = e, for all a ∈ G.

In addition to the above four axioms, if (G, o) satisfies the commutative property, that is, aob=boa for all a,b ∈ G, then the group G is called a commutative group or an abelian group (after the name of Norwegian mathematician Niels Henrik Abel).

For example, (Z, +) is an abelian group as we have m+n=n+m for all m,n ∈ Z.

Examples of Groups

Example 1: (Z, +) is a group, where Z is the set of all integers.

Proof:

1.	We know that for any two integers a and b, we always have a+b∈Z. Thus, the set Z is closed under the operation +.
2.	For any three integers a, b, and c, we have (a+b)+c=a+(b+c). So + is associative on Z.
3.	We have a+0=a for all a∈Z. Thus, 0 is the identity element in Z. In other words, the identity element exists in Z.
4.	For any a∈Z, a+(-a)=0. So -a is the inverse of a in Z. That is, the inverse exists for every element in Z.

Conclusion:

As the above four conditions are satisfied for (Z, +), we can say that the pair (Z, +) is a group. It is called the additive group Z.

More Examples of Groups:

- (R, +) is a group, where R is the set of all real numbers.
- (C, +) is a group, where R is the set of all complex numbers.

Non-Examples of Groups

1	Z is not a group under multiplication, that is, (Z, ×) is not a group. This is because 2 does not have an inverse in Z as 1/2 ∉ Z.
2	The set of real numbers R is not a group under multiplication, that is, (R, ×) is not a group. This is because 0 does not have an inverse in R. Note that (R-{0}, ×) is a group with the identity 1.

Order of Groups

Let (G, o) be a group. The order of the group G is the cardinality of G, denoted by |G|. If |G| is finite, we say that (G, o) is a finite group. Otherwise, it is called an infinite group.

- (Z, +) is an infinite group as the number of elements of Z is not finite.
- (Z/2Z, +) is a finite group of order 2.

Types of Groups

There are many types of groups. For example,

- Abelian Group: commutative groups are known as abelian groups
- Cyclic Group: If a group is generated by a single element, then it is called a cyclic group. For example, (Z, +) is a cyclic group generated by 1.
- Prime order groups: If a group has prime order, then it is called a group of prime order. These groups are always cyclic.

Properties of Groups

Every group (G, o) satisfies the following properties:

1. The composition of two elements always belongs to G. That is, aob∈G for all a,b in G.
2. Every group (G, 0) contains only one identity element.
3. Each element in a group (G, o) has only one inverse.
4. In a group (G, o), the cancellation law holds.
   - aob=aoc ⇒b=c (left cancellation law)
   - boa=coa ⇒b=c (right cancellation law)
5. We have (aob)-1 = b-1oa-1 for all a,b ∈G. That is, the inverse of ab is equal to b-1a-1.

Applications of Group Theory

Group theory has many applications in Physics, Chemistry, Mathematics, and many other areas. It plays a crucial role in public-key cryptography.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

CYCLIC GROUP

LECTURE NO: 38

Cyclic Group

Definition

A group (G, ◦) is called a cyclic group if there exists an element a∈G such that G is generated by a. In other words,

G = {an : n ∈ Z}.

The element a is called the generator of G. Mathematically, it is written as follows: G=<a>. If G is an additive cyclic group that is generated by a, then we have G = {na : n ∈ Z}. The following are a few examples of cyclic groups.

- (Z, +) is a cyclic group. Its generators are 1 and -1.
- (Z4, +) is a cyclic group generated by 1^{-}. It is also generated by 3^{-}.

Non-example of cyclic groups: Klein's 4-group is a group of order 4. It is not a cyclic group.

Order of a Cyclic Group

Let (G, ◦) be a cyclic group generated by a. The order of group G is equal to the order of the element a in G. In other words, |□|=|□|, where |□| denotes the order of the element g. Depending upon whether the group G is finite or infinite, we say G to be a finite cyclic group or an infinite cyclic group.

In the above example, (Z4, +) is a finite cyclic group of order 4, and the group (Z, +) is an infinite cyclic group.

Properties of Cyclic Groups

- If a cyclic group is generated by a, then it is also generated by a-1.
- Every cyclic group is abelian (commutative).
- If a cyclic group is generated by a, then both the orders of G and a are the same.
- Let G be a finite group of order n. If G is cyclic then there exists an element b in G such that the order of b is n.
- Let G be a finite cyclic group of order n and G=<a>. Then G=<ar> if and only if r<n and gcd(r, n)=1. Thus the number of generators of a finite cyclic group of order n is Φ(n), where Φ is the Euler-Phi function.
- Every subgroup of a cyclic group is also cyclic.
- A cyclic group of prime order has no proper non-trivial subgroup.

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

- Let G be a cyclic group of order n. Then G has one and only one subgroup of order d for every positive divisor d of n.
- If an infinite cyclic group G is generated by a, then a and a-1 are the only generators of G.

Problems and Solutions on Cyclic Groups

Question 1: Find all subgroups of the group (Z, +).

Answer:

We know that (Z, +) is a cyclic group generated by 1. As every subgroup of a cyclic group is also cyclic, we deduce that every subgroup of (Z, +) is cyclic, and they will be generated by different elements of Z.

The cyclic subgroup generated by the integer m is (mZ, +), where mZ={mn: n ∈ Z}. As (mZ,+) = (-mZ, +) we conclude that all the subgroups of the group (Z, +) are given as follows:

{(mZ, +) : m is a non-negative integer}.

Question 2: Prove that (Q, +) is not a cyclic group.

Answer:

For a contradiction, we assume that (Q, +) is a cyclic group generated by a. Since (Q, +) is an additive group, we can write it as Q={na : n∈ Z}. As 12a ∈ Q and see that 12a does not belong to the set Q={na : n∈ Z}. Thus a cannot a generator of (Q, +). As a result, the additive group Q is non-cyclic.

Question 3: Prove that (R, +) is not a cyclic group.

Answer:

If possible, we assume that (R, +) is a cyclic group. Note that (Q, +) is a subgroup of (R, +). Thus being a subgroup of a cyclic group, we obtain that (Q, +) is a cyclic group. Thus we arrive at a contradiction because (Q, +) is non-cyclic follows from the above question. So the additive group of real numbers is not a cyclic group.

Question	**Answer**
Is Z cyclic?	As any integer can be written as n= 1+1+…+1 (n times), we see that (Z, +) is generated by 1. Hence, Z is a cyclic group.
Give an example of a non-cyclic group of order 4.	The group Z2 × Z2 has order 4 and it has no element of order 4. So Z2 × Z2 is a noncyclic group of order 4.

Give an example of a cyclic group of order 4.	Z4 is a cyclic group of order 4 generated by 1—.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

MITRC

UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

COSETS

LECTURE NO: 39

Cosets

Definition

A coset of a subgroup H of a group (G, o) is a subset of G obtained by multiplying H with elements of G from left or right.

For example, take H=(Z, +) and G=(Z, +). Then 2+Z, Z+6 are cosets of H in G.

Depending upon the multiplication from left or right we can classify cosets as left cosets or right cosets as follows:

Definition of Left Cosets

Let G be a group and H be a subgroup of G. Then for an element $g \in G$, the left coset of H in G is defined by

$gH = \{gh : h \in H\}$.

The set gH is called a left coset of H in G, and the element g is called a representative of the left coset gH.

Definition of Right Cosets

Let H be a subgroup of a group G. Then the set defined below

$Hg = \{hg : h \in H\}$ where $g \in G$,

is called a right coset of H in G. The element g is called a representative of the right coset Hg.

Examples of Left Cosets and Right Cosets

Let G=(Z, +) and H=(2Z, +). The following are examples of the left cosets of H in G.

- $0+H = \{2n : n \in Z\} = H$
- $1+H = \{2n+1 : n \in Z\}$
- $2+H = \{2n+2 : n \in Z\}$

Below are examples of the right cosets of H in G.

- $H+0 = \{2n : n \in Z\} = H$

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

- H+1 = {2n+1 : n ∈ Z}
- H+2 = {2n+2 : n ∈ Z}

Properties of Cosets

Let G be a group and H be its subgroup. The following are a few properties of left cosets and right cosets.

1. For h ∈ H, the corresponding left (or right) coset is H, that is, hH=H=Hh.
2. H itself a left coset (or a right coset).
3. For h ∉ H, the coset hH and H are distinct.
4. Two cosets are either identical or disjoint.
5. Any two left (or right) cosets have the same cardinality.
6. For a fixed subgroup H of G, the left cosets form a partition of G.
7. Normal subgroups are defined using the concept of cosets. For a normal subgroup N of G, the set of all left cosets of N in G form a group, called the quotient group and it is denoted by G/N.

Applications of Cosets

- Cosets play a crucial role in computational group theory, for example, to calculate the index of a subgroup.
- Cosets are an important tool to prove Lagrange's theorem for finite groups.
- Vitali sets are not measurable which are constructed using cosets.
- Thistlethwaite's algorithm is used to solve Rubik's Cube. It is purely based on the theory of cosets

Theorems on Cosets

Let H be a subgroup of G. Then we have the following theorems on cosets.

Theorem 1: hH=H for h ∈ H.

Proof:

Theorem 2: aH ∩ H = ∅ for a ∈ G-H.

Proof:

Theorem 3: Two cosets are either identical or disjoint, that is,

aH=bH or aH ∩ bH = ∅ .

Proof:

Theorem 4: (Equality of Cosets)

aH=bH if and only if a-1b ∈ H.

Proof:

Theorem 5: (Cardinality of Cosets) Any two left cosets of H in G have the same cardinality, that is,

$|aH|=|bH|$

Proof:

Double Cosets

Let G be a group and H, K be its two subgroups. Then form some $g \in G$, the set

$HgK = \{hgk : h \in H, k \in K\}$

is called a double coset of H and K in G.

Properties of Double Cosets

1. If H=1, then the double cosets are actually left cosets of K in G. Similarly, if K=1, then the double cosets are right cosets of H in G.
2. Two double cosets are either identical or disjoint. In other words, we have either $HxK = HyK$ or $HxK \cap HyK = \varnothing$ for $x, y \in G$.

For fixed subgroups H and K of G, the set of all double cosets form a partition of G.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

NORMAL SUBGROUP AND QUOTIENT GROUP

LECTURE NO: 40

Noromal Subgroup and Quotient Group

Normal subgroup

A **normal subgroup** H of a group G is a subgroup of G which satisfies the similarity transformation with any fixed arbitrary element in G. if G is an abelian group and x is an arbitrary element of G, then Hx is a right coset of H in G and xH is a left coset of in G. Since G is abelian then xH = Hx. However, it is possible that even though G is not an abelian group, it satisfies xH = Hx. In this case subgroups of G are said to be normal subgroups or invariant subgroups or distinguished subgroups.

Definition

Let H be a subgroup of G, then H is said to be a normal subgroup of G, if for every x in G and for h in H xh = xh, that is, xhx-1 belongs to H.

Now since the above statement is true for all h in H. Therefore, we can have

xHx-1 = {xhx-1: for all h in H}, thus normal subgroups of a group G can be defined as:

A subgroup H of a group G is a normal subgroup ⇔ xHx-1 ⊆ H for every x G, where x may or may not be in H.

Normal subgroups are sometimes also referred to as self-conjugates. Normal subgroups are denoted as H ◁ G, it is read as “H is a normal subgroup of G”.

Examples of Normal Subgroup

Every group has necessarily two trivial normal subgroups, viz., the single identity element of G and G itself.

- Let e be the identity element in G, then {e} will be a trivial subgroup of G. Now for every g in G, there exist g-1 in G, then

geg-1 = gg-1 = e ∈ {e}

Thus {e} is the normal subgroup of G.

- Since G is closed with respect to multiplication of its elements, let g, h be any two elements of G, then

ghg-1 = k which could be any element in G. Thus, G itself is a normal subgroup.

Another example of a normal subgroup could be the centre of the group. Let Z be the centre of the group G such that

Z(G) = { z ∈ G | for every g in G, gz = zg}, that elements of Z commute with every element of G.

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

Clearly, gzg-1 = gg-1z = ez = z ∈ Z; where e is an identity element in G.

Thus, Z is a normal subgroup of G.

Properties of a Normal Subgroup

- The intersection of any two normal subgroups of a group is a normal subgroup.
- The intersection of any collection of normal subgroups is a normal subgroup. That is,

If N1, N2, …., Nr are normal subgroups of a group G, then N1 ∩ N2 ∩, ….,∩ Nr is also a normal subgroup of G.

- Every abelian group has a normal subgroup.
- Any group which do not have any normal subgroup other than the trivial normal subgroup is called a simple group.
- If a subgroup is of index 2 in G, that is has only two distinct left or right cosets in G, then H is a normal subgroup of G.
- Every subgroup of a cyclic group is normal.

Solved Examples on Normal Subgroup

Example 1:

Prove that a subgroup H of a group G is a normal subgroup if and only if g-1Hg = H for every g in G.

Solution:

We first take for H be a subgroup of H, g-1Hg = H for every g in G. Then

gHg-1 = H for every g in G

⇒ gHg-1 ⊆ H for every g in G

Hence, by the definition of normal subgroup H is a normal subgroup of G.

Conversely, let H be a normal subgroup of G, then

gHg-1 ⊆ H for every g in G

⇒ H ⊆ g-1Hg for every g in G ……(1)

Now, H ⊆ (g-1)-1Hg-1 for every g in G

⇒ H ⊆ gHg-1 for every g in G

⇒ g-1 H g ⊆ g-1(gHg-1)g for every g in G

⇒ g-1 H g ⊆ (g-1g)H(g-1g) for every g in G

⇒ g-1 H g ⊆ eHe; e is identity in G

⇒ g-1 H g ⊆ H for every g in G ……(2)

From (1) and (2), if H is a normal subgroup of G then g-1Hg = H for every g in G.

Example 2:

Prove that all abelian groups have normal subgroups.

Solution:

Let G be an abelian group and H be a subgroup of G. Since G is abelian therefore all elements of G commutative with respect to multiplication. Let g ∈ G and h ∈ H as H is a subgroup of G the h must belong to G also, then

hg = gh for all g in G

⇒ ghg-1 = h ∈ H for all g in G

⇒ gHg-1⊆ H for all g in G

Hence, every abelian group has normal subgroups.

Quotient Group

Definition

If G is a group and N is a normal subgroup of G, then the set G/N of all cosets of N in G is a group with respect to the composition of the group.

The Quotient Group of G = G/N = { Na | a is in G}

Thus, the elements of G/N are in form of cosets like Na1, Na2, Na3, …where a1, a2, a3,… are in G.

Let Nx and Ny be in G/N then we shall prove that (Nx)(Ny) is in G/N

(Nx)(Ny) = NxyN (since N is a normal subgroup.

Let, xy = z in G, as both x and y belong to G.

(Nx)(Ny) = NzN = NNz = Nz ∈ G/N (since NN = N)

Hence, G/N is closed with respect to the multiplication composition of the group.

Let Na, Nb, Nc are in G/N where a, b and c are in G. Then,

[(Na)(Nb)]Nc = N(ab)c = Na(bc) {since associativity holds within G}

= (Na)N(bc) = Na[N(b)N(c)]

Thus, G/N is associative as well.

Since N = Ne ∈ G/N, for any element Na in G/N, we have

(Na)N = (Na)(Ne) = N(ae) = Na = N(ea) = (Ne)(Na) = N(Na)

Where e is the identity element in G

Hence, N is an identity element in G/N.

If a is an element of G then a-1 is also an element of G, hence Na and Na-1 are elements of G/N such that

(Na)(Na-1) = N(aa-1) = Ne = N

Similarly, (Na-1)(Na) = N.

Thus, for every element in G/N there exists an inverse of it in G/N.

With this, we prove that G/N is a group distinctly called the Quotient Group of G.

Example of a Quotient Group

Let G be the addition modulo group of 6, then G = {0, 1, 2, 3, 4, 5} and N = {0, 2} is a normal subgroup of G since G is an abelian group.

Now, G/N = { N+a | a is in G} = {{0, 2}, {1, 3}, {2, 4}} = { N+0, N+1, N+2}

N + 3 = {3, 5} which will be the same as N + 0.

Let us check whether G/N is group

Closure Property: N+1 and N+2 in G/N

(N + 1)+(N + 2) = N + N + (1 + 2) = N + 3 = N + 0 ∈ G/N

Thus G/N is closed with respect to addition modulo 6.

Associativity: (N+0)[(N+1)+(N+2)] = N + (0+1+2) = N + 3 ∈ G/N

[(N+0)(N+1)]+(N+2) = N + (0+1+2) = N + 3 = N + 0 ∈ G/N

Thus G/N is associative with respect to addition modulo 6.

Identity: Now N +(N + 2) = (N + 0) + (N + 2) = N + 2 = (N + 2) + (N + 0) = (N + 2) + N

Therefore N = N + 0 ∈ G/N is the identity element in G/N.

Inverse: Now, (N + 1)+(N + 2) = N + 3 = N + 0 = (N + 2)+(N + 1)

Thus, N+1 and N+2 are inverse elements of each other in G/N

Therefore, G/N is a Quotient group of G, which is also abelian, as G/N is commutative as well.

Properties of a Quotient Group

- The identity element of a quotient group is the normal subgroup itself.
- If N is a normal subgroup of G, the Order of G/N is equal to the order of G divided by the order of N.That is, |G/H| = |G|/|N|
- Quotient group of an abelian group is abelian, but the converse is not true.
- Every quotient group of a cyclic group is cyclic, but the opposite is not true.
- The quotient group G/G has correspondence to the trivial group, that is, a group with one element.
- The quotient group G/{e} has correspondence to the group itself.
- If G is nilpotent then so is the quotient group G/N.
- If G is solvable then the quotient group G/N is as well.

Solved Examples on Quotient Group

Example 1:

Let G be the additive group of integers and N be the subgroup of G containing all the multiples of 3. The quotient group of G is given by G/N = { N + a | a is in G}. Find the order of G/N.

Solution:

Given G = {…-2, -1, 0, 1, 2, 3,…}

And N = {…, -6, -3, 0, 3, 6,…}

G/N = { N + a | a is in G}

then , N + 1 = {…, -5, -2, -1, 2, 5,…}

N + 2 = {…, -4, -1, 2, 5, 8,…}

Now a = 3b + c where b is in G and c = 0, 1, 2.

Therefore, N + a = N + (3b + c) = (N + 3b) + c = N + c

As 3b belong to N.

Thus, G/N = { N, N + 1, N + 2}

Now, Order of G/N = Index of N in G = Number of cosets of N in G = 3.

Example 2:

Let G = {1, -1, i, -i} be a multiplicative group and N = { 1, -1} be a subgroup of G. Find the number of elements in the quotient group of G.

Solution:

Clearly, G is abelian being a multiplicative group, then N is a normal subgroup.

The quotient group G/N = {Na | a is G}

N1 = { 1, -1} = N

N(-1) = { -1, 1} = N

Ni = { i, -i}

N(-i) = {-i, i} = Ni

G/N = {N, Ni}

Therefore, the number of elements in the quotient group of G is 2.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

HOMOMORPHISM AND ISOMORPHISM

LECTURE NO: 41

Homo-morphism and Isomorphism

Definition of Group Homomorphism

A map $\Phi: G \to G'$ between two groups $(G, 0)$ and $(G', *)$ is called a group homomorphism if the group operation is preserved in the following sense:

$\Phi(a \circ b)=\Phi(a)*\Phi(b) \ \forall \ a,b \in G$

Example of Group Homomorphism

The following is an example of a group homomorphism. The map $\theta: (Z, +) \to (Z, +)$ defined by

$\theta(n)=2n \ \forall \ n \in Z$

is a group homomorphism, because

$\theta(n1+n2)=2(n1+n2) = 2n1+2n2 = \theta(n1)+\theta(n2) \ \forall \ n1, n2 \in Z$

One-to-One homomorphism:

A group homomorphism $\Phi: G \to G'$ is said to be one-to-one (or into) if the map Φ is one-to-one. In other words, Φ is one-to-one if the following holds:

$a=b$ if and only if $\Phi(a)=\Phi(b)$ where $a, b \in G$.

The above map θ is an example of into homomorphism as $\theta(n1)=\theta(n2) \Leftrightarrow 2n1=2n2 = n1=n2$.

Onto homomorphism:

A group homomorphism $\Phi: G \to G'$ is called onto (or surjective) if the map Φ is onto. That is, every element of G has a preimage under the map Φ. It means that for any $g' \in G'$ we have some $g \in G$ such that $\Phi(g)=g'$.

The above map θ is an example of onto homomorphism. This is because for any even integer $2n \in Z$ we have $n \in Z$ such that $\theta(n)=2n$.

Properties of Group Homomorphism

- A one-to-one group homomorphism is called a monomorphism.

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

- An onto group homomorphism is called an epimorphism.
- A group homomorphism is called an isomorphism if it is both one-to-one and onto.
- An isomorphism from a group G onto itself is called an automorphism.

Theorems of Group Homomorphism

Let (G, 0) and (G', *) be two groups and let Φ: G → G' be a group homomorphism.

Theorem 1: Φ(eG) = eG'

That is, a group homomorphism maps identity to identity.

Proof:

We know that eG◦ eG = eG in G. This implies that Φ(eG◦ eG) = Φ(eG). Since Φ is a homomorphism, we have that

Φ(eG) * Φ(eG) = Φ(eG)

⇒ Φ(eG) * Φ(eG) = Φ(eG) * eG'

⇒ Φ(eG) = eG' by the left cancellation law. proved.

Theorem 2: Φ(a-1) = {Φ(a)}-1 for all a ∈ G.

Proof:

For a ∈ G, we have a◦ a-1 = eG = a-1◦ a, where a-1 denotes the inverse of a.

⇒ Φ(a) * Φ(a-1) = Φ(eG) = Φ(a-1) * Φ(a) as Φ is a homomorphism.

⇒ Φ(a) * Φ(a-1) = eG' = Φ(a-1) * Φ(a) by Theorem 1.

So by the definition of an inverse, we conclude that Φ(a-1) is the inverse of Φ(a). In other words,

Φ(a-1) = {Φ(a)}-1 proved.

Theorem 3: If a ∈ G and the order of a is finite, then the order of Φ(a) is a divisor of the order of a. In other words,

◦ (�) | ◦ (�(�))

Proof:

Definition of a Kernel of a Homomorphism

Let Φ: (G, 0) → (G', *) be a group homomorphism, where G and G' are two groups. Then the kernel of Φ, denoted by ker(Φ), is defined by

ker(Φ) = {a ∈ G : Φ(a)=eG'}.

Thus, the kernel of a group homomorphism Φ is the set of all elements of G that are mapped to the identity element of G' under the map Φ.

Kernel is a Normal Subgroup

Theorem: The kernel of a homomorphism is a normal subgroup.

Proof:

Let Φ: (G, 0) → (G', *) be a group homomorphism. We will show that ker(Φ) is a normal subgroup of G. The following steps have to follow.

Step 1: As Φ(eG)=eG', we have eG ∈ ker(Φ). Thus, ker(Φ) is a non-empty subset of G.

Step 2: To show ker(Φ) is a subgroup of G.

Let a, b ∈ ker(Φ).

Thus Φ(a) = eG', Φ(b) = eG'

Now since Φ is a homomorphism, we have

Φ(a ◦ b-1) = Φ(a) * Φ(b-1) = eG'*eG' = eG'

⇒ a ◦ b-1 ∈ ker(Φ) by the definition of kernel.

This shows that ker(Φ) is a subgroup of G.

Step 3: To show ker(Φ) is normal in G.

Let g ∈ G and h∈ ker(Φ) ⇒ Φ(h) = eG'.

Now Φ(g ◦ h ◦ g-1) = Φ(g) * Φ(h) * Φ(g-1) as Φ is a group homomorphism.

= Φ(g) * eG' * Φ(g-1) as Φ(h) = eG'

= Φ(g) * Φ(g-1)

= Φ(g ◦ g-1) by the fact that Φ is a homomorphism.

= Φ(eG)=eG'

$\Rightarrow \Phi(g \circ h \circ g^{-1}) = eG'$

$\Rightarrow g \circ h \circ g^{-1} \in \ker(\Phi)\ \forall\ g \in G$ and $h \in \ker(\Phi)$.

Injectivity criteria for homomorphism

Theorem: Let $\Phi: (G, 0) \rightarrow (G', *)$ be a group homomorphism. Then Φ is one-to-one if and only if $\ker(\Phi) = \{eG\}$.

Proof:

First, we assume that Φ is one-to-one. We know that $\Phi(eG)=eG'$. So eG is a preimage of eG' under the map Φ. Since Φ is one-to-one, it is the only preimage of eG'. Therefore, the set $\{a \in G : \Phi(a)=eG'\} = \{eG\}$. In other words, $\ker(\Phi) =\{eG\}$, that is, $\ker(\Phi)$ is trivial.

Next, we assume that $\ker(\Phi) = \{eG\}$. We need to show that Φ is one-to-one.

Let $a, b \in \ker(\Phi)$ be such that $\Phi(a)=\Phi(b)$.

$\Rightarrow \Phi(ab^{-1})=eG'$

$\Rightarrow ab^{-1} \in \ker(\Phi)= \{eG\}$

$\Rightarrow ab^{-1} = eG$

$\Rightarrow a=b.$

Isomorphism of Group :
Let (G,o) & (G',o') be 2 groups, a mapping " f " from a group (G,o) to a group (G',o') is said to be an isomorphism if –

1. f(aob) = f(a) o' f(b) ∀ a,b ∈ G2. f is a one- one mapping3. f is an onto mapping.

If 'f' is an isomorphic mapping, (G,o) will be isomorphic to the group (G',o') & we write :

$G \cong G'$

Note : A mapping f: X -> Y is called :

1. One – One – If $x1 \neq x2$, then $f(x1) \neq f(x2)$ or if $f(x1) = f(x2) \Rightarrow x1 = x2$. Where $x1, x2 \in X$
2. Onto – If every element in the set Y is the f-image of at least one element of set X.
3. Bijective – If it is one & Onto.

Example of Isomorphism Group –
If G is the multiplicative group of 3 cube-root units , i.e., $(G,o) = (\{1, w, w2\}, *)$ where $w3 = 1$ & G' is an additive group of integers modulo 3 – $(G', o') = (\{1,2,3), +3)$. Then : $G \cong G'$, we say G is isomorphic to G'.

First Isomorphism Theorem for Groups

Statement: Let G and G' be two groups. Let $\Phi: G \to G'$ be a homomorphism. Then we have the following isomorphism of groups

$G/\ker(\Phi) \cong \text{Im}(\Phi)$

Proof of First Isomorphism Theorem

The below steps have to be followed to prove the first isomorphism theorem of groups. We will proceed as follows:

Step 1: We know that $\ker(\Phi)$ is a normal subgroup of G. Thus the quotient group $G/\ker(\Phi)$ is well-defined. So we define a map

$\theta: G/\ker(\Phi) \to \text{Im}(\Phi)$

by $\theta(aH) = \Phi(a)$ for all $a \in G$. Write $H=\ker(\Phi)$. So $\theta: G/H \to \text{Im}(\Phi)$.

Let $aH=bH$. We will show that $\theta(aH)=\theta(bH)$.

Since $aH=bH$ we have $a\text{-}1b \in H=\ker(\Phi)$. Thus we have

$\Phi(a\text{-}1b)=eG'$

$\Leftrightarrow \Phi(a)\text{-}1\ \Phi(b) = eG'$

$\Leftrightarrow \Phi(a) = \Phi(b)$

$\Leftrightarrow \theta(aH)=\theta(bH)$.

This shows that the map θ is well-defined and one-to-one.

Step 2: To show the map θ is a homomorphism. Write $aH, bH \in G/H$ for a, b in G.

$\theta(aH * bH) = \theta(abH)$ where $*$ is the composition on the group G/H.

$=\Phi(ab)$

$=\Phi(a)\ \Phi(b)$ as Φ is a homomorphism

$= \theta(aH)\ \theta(bH)$

This shows that the map θ is a group homomorphism.

Step 4: By definition, the map θ is onto. This is because: take $g' \in \text{Im}(\Phi)$; so we have $g \in G$ such that $\Phi(g)=g'$.

Now $\theta(gH) = \Phi(g)=g'$. So g' has a preimage gH under the map θ. Hence, the map θ is onto.

Thus we have shown that the map θ is a one-to-one and onto homomorphism. In other words, θ is a group isomorphism, that is, $G/H=G/\ker(\Phi) \cong \text{Im}(\Phi)$. This completes the proof of the first isomorphism theorem for groups.

Properties of First Isomorphism Theorem

From the first isomorphism theorem of groups, we can conclude the following points:

- If G is a group, then every homomorphic image of G is isomorphic to a quotient group G/H for some normal subgroup H of G.
- In some sense, the natural quotient homomorphisms G→G/H describe all possible homomorphisms of G.
- The total number of homomorphisms of G = The number of different natural quotient homomorphisms G→G/H = The number of different normal subgroups H of G.
- The first isomorphism theorem for groups is also called the fundamental theorem of group homomorphisms.

Application of First Isomorphism Theorem

We will give a few examples as an application of the first isomorphism theorem for groups.

Question 1: Let G be a group of order 12 and G' be a group of order 5. Show that there does not exist a homomorphism from G onto G'.

Solution:

If possible, suppose that $\Phi: G \to G'$ is an onto homomorphism. Then by the first isomorphism of groups, we have that

$G/\ker(\Phi) \cong G'$

Thus both groups $G/\ker(\Phi)$ and G' have the same order. So we get that

$|G/\ker(\Phi)| = |G'|$

$\Rightarrow |G| / |\ker(\Phi)| = |G'|$

$\Rightarrow 12 / |\ker(\Phi)| = 5$

$\Rightarrow |\ker(\Phi)| = 12/5$ which is not a positive integer. Thus we arrive at a contradiction. So there does not exist a group homomorphism from G onto G'.

Second Isomorphism Theorem Statement

Let G be a group such that

1. H be a subgroup of G

2. K be a normal subgroup of G.

Then we have a group isomorphism

$H/(H \cap K) \cong HK/K$.

Second Isomorphism Theorem Proof

The below steps have to be followed to prove the second isomorphism theorem for groups.

Step 1: To show HK is a subgroup of G.

In order to prove this, we need to show that HK=KH. For $h \in H$, $k \in K$, as K is normal in G, we have that $hkh^{-1} \in K$.

$\therefore hk = (hkh^{-1})h \in KH$.

Hence, $HK \subseteq KH$.

In a similar way, $KH \subseteq HK$.

Therefore HK = KH, proving that HK is a subgroup of G.

Step 2: To show K is normal in HK.

As HK is a subgroup of G by Step 1, and K is normal in G by assumption, one can easily deduce that K is a normal subgroup of HK.

Step 3: To show $H \cap K \trianglelefteq H$. This follows from the fact that $K \trianglelefteq G$.

We will now prove the isomorphism.

Step 4: Define a mapping

$\phi: H \rightarrow HK/K$

by $\phi(h) = hK$ for $h \in H$.

Note that $\phi(h_1h_2) = h_1h_2K = (h_1K)(h_2K) = \phi(h_1)\phi(h_2)$, so ϕ is a group homomorphism.

By the definition of ϕ, ϕ is onto.

Now, $\text{Ker } \phi = \{h \in H: \phi(h) = K\}$

$= \{h \in H: hK = K\}$

$= \{h \in H: h \in K\}$

= H∩K.

So by the first isomorphism theorem of groups, we can conclude that

H/(H∩K) ≅ HK/K.

This proves the second isomorphism theorem for groups.

Third Isomorphism Theorem Statement

Let G be a group and H, K be its two normal subgroups such that H ≤ K. Then we have a group isomorphism

(G/H)/(K/H) ≅ G/K.

Third Isomorphism Theorem Proof

First, we prove that K/H is a normal subgroup of G/H. For gH∈ G/H and kH ∈ K/H, we have that

gH kH (gH)-1

= gH kH g-1 H

= gkg-1 H

∈ K/H as gkg-1∈K. This is because K is normal in G.

This shows that K/H is normal in G/H.

Next, we prove the group isomorphism. Let us define a mapping

ϕ: G/H → G/K

by ϕ(gH) = gK for g ∈ G.

Let us show that ϕ is well-defined. For g1, g2∈G with g1H=g2H, we have that

g1 = g2h for some element h∈H.

As H ≤ K, we have h∈K. Thus, we can conclude that

g1K=g2K

⇒ ϕ(g1H) = ϕ(g2H).

This shows that ϕ is well-defined.

By definition, ϕ is onto.

Now, Ker ϕ = {gH∈ G/H: ϕ(gH) = K}

= {gH∈ G/H: gK = K}

= {gH∈ G/H: g ∈ K}

= G/K.

Hence, by the first isomorphism theorem of groups, we obtain that

$G/Ker\,\phi \cong G/K$.

This completes the proof of the third isomorphism theorem for groups.

$+_3$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$*_3$	1	W	W^2
1	1	W	W^2
W	W	W^2	$W^3 = 1$
W^2	W^2	$W^3 = 1$	W4= 1.W = W

- The structure & order of both the tables are same. The mapping 'f' is defined as : f : G -> G' in such a way that f(1) = 0 , f(w) = 1 & f(w2) = 2.
- Homomorphism property : f(aob) = f(a) o' f(b) ∀ a,b ∈ G . Let us take a = w & b = 1
  LHS : f(a * b) = f(w * 1) = f(w) = 1.
  RHS : f(a) +3 f(b) = f(w) +3 f(1) = 1 + 0 = 1
  =>LHS = RHS
- This mapping f is one-one & onto also, therefore, a homomorphism.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-5 (DISCRETE MATHEMATICAL STRUCTURE)

GROUP PERMUTATION

LECTURE NO: 42

Group Permutation

Let G be a non-empty set, then a one-one onto mapping to itself that is as shown below is called a permutation.

$$P: G \xrightarrow[\text{onto}]{\text{one-one}} G$$

- The number of elements in finite set G is called the degree of Permutation.
- Let G have n elements then Pn is called a set of all permutations of degree n.
- Pn is also called the Symmetric group of degree n.
- Pn is also denoted by Sn.
- The number of elements in Pn or Sn is $n!$

Examples:

Case1: *Let G={ 1 } element then permutation are Sn or Pn =* $\begin{pmatrix}1\\1\end{pmatrix}$

Case 2: *Let G= { 1, 2 } elements then permutations are* $\begin{pmatrix}1&2\\1&2\end{pmatrix}, \begin{pmatrix}1&2\\2&1\end{pmatrix}$

Case 3: *Let G={ 1, 2, 3 } elements then permutation are* ***3!=6.*** *These are,*

$$\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix}, \begin{pmatrix}1&2&3\\2&1&3\end{pmatrix}, \begin{pmatrix}1&2&3\\3&2&1\end{pmatrix}, \begin{pmatrix}1&2&3\\1&3&2\end{pmatrix}, \begin{pmatrix}1&2&3\\3&1&2\end{pmatrix}, \begin{pmatrix}1&2&3\\2&3&1\end{pmatrix}$$

Reading the Symbol of Permutation

Suppose that a permutation is $\begin{pmatrix}1&2&3&4&5&6\\2&3&1&4&5&6\end{pmatrix}$

- First, we see that in a small bracket there are two rows written, these two rows have numbers. The smallest number is 1 and the largest number is 6.
- Starting from the left side of the first row we read as an image of 1 is 2, an image of 1 is 2, an image of 2 is 3, an image of 3 is 1, an image of 4 is 4 **(Self image=identical=identity)**, an image of 5 is 6 and image of 6 is 5.
- The above thing can be also read as: Starting from the left side of the first row 1 goes to 2, 2goes to 3, 3goes to,4 goes to 4,5 goes to 6, and 6 goes to 5.

A cycle of length 2 is called a **permutation.**

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

Example:

1) $\begin{pmatrix} 4 & 5 \end{pmatrix}$

Length is 2, so it is a transposition.

2) $\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$

Length is three, so it is not a transposition.

Multiplication of Permutation

Problem:

If

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix} and\, B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}$$

Find the product of permutation A.B and B.A

Solution:

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix} and\, B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}$$

$$A.B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ & & & & \end{pmatrix}$$

Here we can see that in first bracket 1 goes to 2 i.e. image of 1 is 2, and in second row 2 goes to 3 i.e. image of 2 is 3.

Hence, we will write 3 under 1 in the bracket shown below,

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & & & & \end{pmatrix}$$

Do above step with all elements of first row, answer will be

$$A.B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 5 & 2 \end{pmatrix}$$

Similarly,

$$B.A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$$
$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 5 & 3 \end{pmatrix}$$

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

MITRC